

TECHNICAL SPECIFICATION DOCUMENT

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Prepared By: Product Owner

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1. Executive Summary

1.1 Introduction

Subscription-based businesses such as OTT streaming platforms, SaaS providers, online learning platforms, and digital subscription services rely heavily on recurring customer subscriptions for sustainable revenue. Customer retention is therefore one of the most important factors influencing business growth and profitability. However, customers may discontinue their subscriptions due to various reasons such as reduced product usage, dissatisfaction with the service, perceived high subscription costs, or changes in customer preferences. This phenomenon, commonly referred to as **customer churn**, directly affects recurring revenue, customer lifetime value, and overall business performance.

Currently, businesses often identify customer cancellations only after they occur, leaving limited opportunities to engage with customers before they leave. This reactive approach results in higher customer acquisition costs, reduced profitability, and missed opportunities to improve customer relationships.

The purpose of this project is to develop an intelligent **Subscription Cancellation Prediction System** capable of identifying customers who are likely to cancel their subscriptions before the cancellation occurs. By leveraging historical customer information, subscription details, and usage behavior, the system will generate predictive insights that enable businesses to proactively implement retention strategies. In addition to churn prediction, the proposed solution will provide customer risk scores, explainable prediction results, customer segmentation, analytical dashboards, and reporting capabilities. These features will assist business teams in understanding customer behavior, prioritizing retention efforts, and making informed business decisions.

1.2 Purpose of the Document

This Product Requirements Document (PRD) defines the business expectations, functional requirements, product features, project scope, and overall objectives for the proposed Subscription Cancellation Prediction System.

The primary purpose of this document is to communicate the business vision and product expectations to the Technical Team Lead before development begins. The document serves as a reference for solution planning, effort estimation, sprint planning, architecture design, and implementation.

This document focuses on **what the product should achieve** from a business perspective. Detailed technical design decisions, implementation approaches, database structures, APIs, and software architecture will be prepared separately by the Technical Team Lead during the solution design phase.

1.3 Project Vision

The vision of the Subscription Cancellation Prediction System is to provide businesses with an intelligent decision-support platform capable of proactively identifying customers at risk of cancellation and enabling timely customer retention actions.

Rather than functioning solely as a machine learning prediction application, the system is intended to become a comprehensive customer intelligence platform. It will

consolidate customer information, predictive analytics, risk assessment, explainable artificial intelligence, business dashboards, and reporting into a unified web application that supports business users in improving customer retention and reducing revenue loss.

The long-term vision of the product is to assist organizations in transforming customer retention from a reactive process into a proactive, data-driven business strategy.

2. Business Problem

2.1 Background

Subscription-based business models depend on continuous customer renewals to maintain consistent revenue streams. Every customer cancellation results not only in immediate revenue loss but also increases customer acquisition costs, reduces customer lifetime value, and impacts long-term business growth.

Many organizations currently rely on historical reports or manual analysis to identify customers who have already cancelled their subscriptions. This reactive approach limits the organization's ability to intervene before cancellation occurs.

The absence of predictive customer analytics makes it difficult for business teams to determine:

- Which customers are likely to cancel.
- Which customer segments require immediate attention.
- Which factors contribute most to customer churn.
- Which retention strategies should be applied.
- How customer behavior changes over time.

As a result, businesses often allocate retention resources inefficiently and fail to prioritize customers who are at the highest risk of leaving.

2.2 Problem Statement

The organization currently lacks an intelligent system capable of predicting future subscription cancellations using historical customer data.

Without predictive insights, the business faces several operational challenges:

- Inability to identify customers at risk before cancellation.
- Limited understanding of customer behavior leading to churn.
- Difficulty prioritizing customer retention efforts.
- Manual analysis of customer records.
- Lack of centralized customer analytics.
- Limited visibility into customer risk levels.
- Reduced effectiveness of customer engagement strategies.

These challenges contribute directly to increased customer churn and declining recurring revenue.

2.3 Proposed Solution

The proposed solution is a web-based Subscription Cancellation Prediction System that combines machine learning with business analytics to assist the organization in identifying customers who are likely to cancel their subscriptions.

The system will analyze customer information, subscription details, usage behavior, and other available customer attributes to generate churn predictions along with a customer risk score.

In addition to prediction, the system will provide customer profiles, explainable AI insights, customer ranking, segmentation, retention recommendations, dashboards,

analytics, and reporting features. These capabilities will enable business users to make informed decisions and proactively engage with customers before subscription cancellation occurs.

The solution is designed to support business decision-making rather than replace human judgment, providing actionable insights that improve customer retention strategies.

3. Project Objectives

3.1 Primary Objective

The primary objective of the Subscription Cancellation Prediction System is to develop an intelligent web-based application capable of predicting whether a customer is likely to cancel their subscription based on historical customer information, subscription details, and usage behavior.

The system aims to assist the business in identifying customers at risk before cancellation occurs, enabling proactive customer retention strategies that reduce churn and improve recurring revenue.

3.2 Secondary Objectives

In addition to churn prediction, the system will support business users by providing customer intelligence and analytical capabilities that improve decision-making.

The secondary objectives of the project are:

- To provide a customer churn risk score for every customer.
- To explain the key factors contributing to each churn prediction.
- To rank customers based on their churn probability.
- To segment customers into meaningful groups for business analysis.
- To recommend suitable retention strategies for high-risk customers.
- To provide a centralized dashboard displaying business KPIs and customer insights.
- To generate reports that support management review and business planning.
- To provide a searchable customer repository with prediction history and customer information.

3.3 Expected Business Outcomes

The successful implementation of the system is expected to deliver the following business outcomes:

- Early identification of customers likely to cancel their subscriptions.
- Improved customer retention through proactive intervention.
- Reduced revenue loss caused by customer churn.
- Improved visibility into customer behavior and churn trends.
- Better allocation of customer success resources by prioritizing high-risk customers.
- Data-driven business decisions through analytical dashboards and reports.
- Increased operational efficiency by automating customer churn analysis.

4. Project Scope

The scope of the project defines the functionalities that will be included in the current version of the Subscription Cancellation Prediction System.

4.1 In Scope

The following functionalities are included within the scope of the project:

User Authentication

- Secure login for authorized users.
- Session management.

Admin access to all system modules.

- View customer records.
- Search customers.
- Filter customer information.
- View detailed customer profiles.

Churn Prediction

- Predict whether a customer is likely to cancel.
- Display prediction probability.
- Display customer churn risk score.
- Display prediction result (Likely to Cancel / Not Likely to Cancel).

Customer Intelligence

- Customer ranking based on churn probability.
- Customer segmentation.
- Explainable AI insights.
- Customer risk analysis.

Retention Recommendation Engine

- Display appropriate retention recommendations for customers identified as high risk.

Examples include:

- Discount offers
- Promotional offers
- Free trial recommendations
- Customer support follow-up

Dashboard

Display business KPIs including:

- Total customers
- Predicted churn customers
- Average churn risk
- Average customer satisfaction
- Monthly spending overview
- Customer segmentation overview

Analytics

Provide interactive visualizations including:

- Customer distribution
- Spending analysis
- Satisfaction analysis
- Usage analysis
- Device type analysis

- Payment mode analysis
- Churn distribution

Reports

Generate downloadable reports including:

- High-risk customers
- Customer summaries
- Prediction summaries
- Segmentation reports

Model Evaluation

Display machine learning performance metrics including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix
- Feature Importance

4.2 Out of Scope

The following functionalities are not included in the current project scope:

- Mobile application
- Real-time customer activity monitoring
- CRM integration
- Email marketing automation
- SMS notifications
- Automatic model retraining
- Real-time streaming analytics
- Cloud-based multi-tenant deployment
- Customer self-service portal

These functionalities may be considered for future releases.

5. Stakeholders

The successful development and implementation of the Subscription Cancellation Prediction System requires collaboration among multiple stakeholders. Each stakeholder has distinct responsibilities throughout the project lifecycle.

Stakeholder	Responsibility
Business Owner	Defines business goals, approves product requirements, and validates business outcomes.
Product Owner	Defines product vision, gathers requirements, prioritizes features, prepares the Product Requirements Document, and coordinates with the technical team.
Technical Team Lead	Reviews requirements, prepares the technical solution, estimates effort, assigns development tasks, and oversees implementation.
Machine Learning Engineer	Develops, trains, evaluates, and optimizes the churn prediction model.

Stakeholder	Responsibility
Backend Developer	Implements APIs, business logic, authentication, and database integration.
Frontend Developer	Develops the web interface, dashboards, analytics pages, and user interactions.
QA Engineer	Validates functional and non-functional requirements through testing.
Business Analyst (Future)	Uses dashboards and reports to analyze customer behavior and churn trends.
Customer Success Executive (Future)	Uses prediction results and recommendations to engage with high-risk customers.

6. User Roles

The current version of the application will support a single user role: **Administrator**. Future versions of the system may introduce additional roles such as Business Analyst and Customer Success Executive.

6.1 Administrator

The Administrator is the primary user of the Subscription Cancellation Prediction System.

The Administrator is responsible for managing customer information, monitoring churn predictions, analyzing customer behavior, and generating business reports.

Responsibilities

The Administrator shall be able to:

- Access the application through secure authentication.
- View the executive dashboard.
- Search and view customer records.
- Access individual customer profiles.
- View churn predictions and risk scores.
- View prediction explanations.
- Analyze customer segments.
- View customer rankings.
- Perform bulk churn prediction.
- Generate business reports.
- View machine learning model performance metrics.

Permissions

The Administrator shall have permission to:

- Access all modules within the application.
- Upload customer datasets for bulk prediction.
- Export prediction reports.
- View analytical dashboards.
- View model evaluation metrics.

Future User Roles

The following roles are identified for future versions of the system:

Business Analyst

- View dashboards
- Analyze customer trends
- Generate reports
- Access analytical insights

Customer Success Executive

- View high-risk customers
- Review churn reasons
- View retention recommendations
- Monitor customer follow-up activities

7. Business Requirements

7.1 Overview

The Subscription Cancellation Prediction System is intended to provide businesses with an intelligent platform for identifying customers who are likely to cancel their subscriptions before the cancellation occurs. The system should assist business users by providing predictive insights, customer analytics, and decision-support tools that improve customer retention and reduce revenue loss.

The business requires a centralized web-based application that combines customer information, predictive analytics, explainable artificial intelligence, reporting, and visualization capabilities into a single platform.

The following business requirements have been identified based on discussions with the business owner.

7.2 Business Requirements

BR-001 – Customer Churn Prediction

The system shall predict whether a customer is likely to cancel their subscription using the available customer and subscription data.

The prediction shall classify customers into one of the following categories:

- Likely to Cancel
- Not Likely to Cancel

The prediction shall also include a probability score indicating the confidence of the prediction.

Priority: High

BR-002 – Customer Risk Assessment

The system shall assign a churn risk score to every customer.

The risk score shall help business users understand the likelihood of customer cancellation and prioritize customer retention activities.

The risk score shall be displayed as a percentage ranging from 0% to 100%.

Priority: High

BR-003 – Customer Information Management

The system shall provide access to customer information, allowing users to search, filter, and view customer profiles.

Customer information shall include customer demographics, subscription details, usage information, and prediction results.

Priority: High

BR-004 – Customer Intelligence

The system shall provide customer intelligence features that enable business users to understand customer behavior through customer ranking, customer segmentation, and prediction explanations.

These insights shall support better business decision-making and customer prioritization.

Priority: High

BR-005 – Retention Support

The system shall recommend appropriate customer retention strategies for customers identified as having a high probability of cancellation.

Recommendations shall assist business users in selecting suitable actions for reducing customer churn.

Priority: High

BR-006 – Business Dashboard

The system shall provide an executive dashboard displaying important business metrics related to customer churn and customer behavior.

The dashboard shall provide a consolidated overview of customer health and business performance.

Priority: High

BR-007 – Business Analytics

The system shall provide analytical visualizations that help business users understand customer trends, spending behavior, satisfaction levels, usage patterns, and churn distribution.

The analytics shall support data-driven business decisions.

Priority: High

BR-008 – Bulk Prediction

The system shall support prediction of multiple customer records through CSV upload.

The prediction results shall be downloadable for further analysis.

Priority: Medium

BR-009 – Reporting

The system shall generate business reports containing prediction summaries, customer rankings, customer segmentation, and analytical information.

Reports shall support business review meetings and decision-making processes.

Priority: High

BR-010 – Model Transparency

The system shall provide model evaluation metrics that allow business users to understand the overall performance and reliability of the prediction model.

Priority: Medium

8. Product Features

8.1 Feature Overview

The Subscription Cancellation Prediction System consists of ten primary product features designed to address the business requirements identified during requirement gathering.

Each feature contributes towards achieving the overall objective of reducing customer churn through predictive analytics and customer intelligence.

The following sections describe each feature in detail.

Feature FTR-001

Churn Prediction

Description

The Churn Prediction feature is the core functionality of the system. It analyzes customer information and predicts whether a customer is likely to cancel their subscription in the near future.

The prediction is generated using a trained machine learning model based on the available dataset.

The prediction result shall be presented in a simple and understandable format for business users.

Business Value

This feature enables the business to proactively identify customers who are at risk of leaving before the cancellation occurs. Early identification allows the customer success team to implement suitable retention strategies and reduce customer churn.

Inputs

The prediction shall use the available customer attributes, including:

- Age
- Income Level
- Number of Subscriptions
- Tenure
- Monthly Total Spend
- Average Usage Hours
- App Switch Frequency
- Customer Support Interactions
- Satisfaction Score
- Discount Used
- Device Type
- Payment Mode

Outputs

The system shall display:

- Prediction Result
- Churn Probability
- Risk Category

Example:

Prediction

Likely to Cancel

Probability
89%
Risk Category

Acceptance Criteria

The feature shall be considered complete when:

- The system predicts churn successfully.
- Prediction probability is displayed.
- Prediction result is understandable.
- Prediction completes within acceptable response time.

Feature FTR-002

Customer Profile & Churn Insights

Description

The Customer Profile feature provides a comprehensive view of an individual customer.

The page combines customer information, churn prediction, customer risk score, and prediction explanations into a single interface.

This feature enables business users to understand both customer characteristics and churn behavior without navigating across multiple screens.

Business Value

The Customer Profile serves as a centralized customer intelligence page that assists business users in making informed retention decisions.

Instead of viewing raw customer records, users receive meaningful insights explaining why a customer is considered high risk.

Customer Information

The profile shall display:

- Customer ID
- Age
- Income Level
- Number of Subscriptions
- Customer Tenure
- Monthly Spend
- Usage Hours
- Device Type
- Payment Mode
- Satisfaction Score
- Customer Support Interactions

Churn Insights

The page shall additionally display:

- Prediction Result
- Risk Score
- Risk Category
- Explainable AI Insights
- Top Factors Influencing Prediction

Business Value of Explainability

Providing explanation for every prediction improves business confidence in the model and helps customer success teams understand the factors contributing to customer churn.

Acceptance Criteria

The feature shall be considered complete when:

- Customer profile is displayed successfully.
- Customer information is clearly organized.
- Prediction is displayed.
- Risk score is displayed.
- Prediction explanation is displayed.
- Customer details load successfully.

Feature FTR-003

Customer Ranking & Segmentation

Description

The Customer Ranking & Segmentation feature enables business users to identify, prioritize, and categorize customers based on their predicted churn risk and behavioral characteristics. The feature combines churn prediction results with customer segmentation to provide a structured view of the customer base.

Customer Ranking orders customers according to their churn probability, allowing business teams to focus on customers who require immediate attention. Customer Segmentation groups customers with similar characteristics to support targeted retention strategies and business analysis.

Business Value

Customer retention resources are often limited. By ranking customers according to churn risk, the business can prioritize intervention efforts where they are most needed. Customer segmentation provides additional business insights by identifying groups of customers with similar characteristics, enabling personalized business strategies instead of applying the same retention approach to every customer.

Functional Capabilities

The system shall:

- Rank customers based on churn probability.
- Display customer risk category.
- Group customers into meaningful segments.
- Allow users to view customers within each segment.
- Update rankings automatically after prediction.

Customer Segments

The system shall classify customers into logical business segments based on customer characteristics.

Example segments include:

- High Risk Customers
- Loyal Customers
- Premium Customers
- Budget Customers

The exact segmentation logic may be refined during implementation based on the available dataset.

Outputs

The feature shall display:

- Ranked Customer List
- Customer Segment
- Risk Score
- Prediction Status

Acceptance Criteria

The feature shall be considered complete when:

- Customers are ranked based on churn probability.
- Customer segments are displayed.
- Ranking updates automatically after prediction.
- Users can filter customers by segment.

Feature FTR-004

Retention Recommendation Engine

Description

The Retention Recommendation Engine provides business users with suggested actions that may reduce the likelihood of customer churn. Recommendations are generated using predefined business rules based on customer characteristics and prediction results.

The recommendations are intended to support customer success teams in selecting appropriate engagement strategies for high-risk customers.

Business Value

Predicting customer churn alone does not reduce cancellations. The business also requires actionable recommendations that assist customer success teams in engaging customers before they leave.

The Retention Recommendation Engine transforms prediction results into practical business actions.

Functional Capabilities

The system shall:

- Display retention recommendations for high-risk customers.
- Generate recommendations based on customer behavior.
- Display recommendations within the customer profile.
-

Example Recommendations

Depending on customer characteristics, the system may recommend:

- Offer Discount
- Provide Free Trial
- Contact Customer Support
- Recommend Subscription Upgrade
- Provide Personalized Offers

Outputs

The feature shall display:

- Recommendation Type
- Recommendation Description

Acceptance Criteria

The feature shall be considered complete when:

- Recommendations are displayed for customers.
- Recommendations are generated consistently.
- Recommendations are visible within the Customer Profile.

Feature FTR-005

Executive Dashboard (KPIs)

Description

The Executive Dashboard serves as the landing page of the application and provides business users with a consolidated overview of customer churn and business performance.

The dashboard shall present key performance indicators (KPIs) that summarize customer health and prediction results.

Business Value

The Executive Dashboard allows business users to quickly understand the overall health of the customer base without reviewing individual customer records.

The dashboard supports faster decision-making through visual summaries.

Dashboard KPIs

The dashboard shall display:

- Total Customers
- Predicted Churn Customers
- High Risk Customers
- Average Churn Risk Score
- Average Customer Satisfaction Score
- Average Monthly Spend
- Average Customer Tenure

Acceptance Criteria

The feature shall be considered complete when:

- Dashboard loads successfully.
- KPIs are calculated correctly.
- Dashboard updates when prediction results change.
- Information is presented in an easy-to-understand format.

Feature FTR-006

Analytics Dashboard

Description

The Analytics Dashboard provides interactive visualizations that help business users understand customer behavior and identify patterns related to customer churn.

The dashboard transforms raw customer information into meaningful visual insights.

Business Value

Visual analytics simplify business analysis by allowing users to identify patterns and trends that may not be immediately apparent from tabular data.

The dashboard supports strategic decision-making by providing graphical representations of customer behavior.

Functional Capabilities

The Analytics Dashboard shall provide:

- Interactive Charts
- Filtering
- Drill-down capabilities (future enhancement)
- Summary statistics

Proposed Charts

The dashboard may include visualizations such as:

- Customer Distribution by Income Level
- Device Type Distribution
- Payment Mode Distribution
- Monthly Spend Distribution
- Satisfaction Score Distribution
- Customer Tenure Distribution
- Churn Distribution
- Customer Segmentation Distribution

Acceptance Criteria

The feature shall be considered complete when:

- Charts load correctly.
- Visualizations represent available data accurately.
- Users can interact with the charts.
- Filters update the displayed information.

Feature FTR-007

Customer Search & Filter

Description

The Customer Search & Filter feature enables users to quickly locate customer records and analyze specific groups of customers.

The feature improves usability by reducing the time required to access customer information.

Business Value

Businesses managing thousands of customer records require efficient search and filtering capabilities.

This feature improves operational efficiency by enabling users to locate customers based on business-specific criteria.

Functional Capabilities

The system shall allow users to:

- Search by Customer ID.
- Filter by Income Level.
- Filter by Payment Mode.

- Filter by Device Type.
- Filter by Risk Category.
- Filter by Prediction Status.

Acceptance Criteria

The feature shall be considered complete when:

- Customer search returns matching records.
- Filters display correct results.
- Multiple filters can be applied simultaneously.

Feature FTR-008

Bulk Prediction

Description

The Bulk Prediction feature enables users to predict subscription cancellation for multiple customers simultaneously by uploading a CSV file containing customer records. This feature is designed to eliminate the need for predicting each customer individually and significantly improve operational efficiency when dealing with large datasets.

After processing the uploaded dataset, the system shall generate predictions for all valid customer records and provide the results in a downloadable format.

Business Value

Organizations often manage thousands of customer records. Performing individual predictions for each customer is time-consuming and inefficient.

The Bulk Prediction feature enables businesses to process an entire customer dataset in a single operation, making large-scale churn analysis practical and efficient.

Functional Capabilities

The system shall:

- Allow users to upload customer data in CSV format.
- Validate the uploaded dataset before processing.
- Perform churn prediction for all valid customer records.
- Generate churn probability and risk score for each customer.
- Display prediction progress.
- Allow users to download prediction results.

Inputs

The uploaded CSV file shall contain customer attributes required by the prediction model.

The uploaded data shall follow the predefined dataset format.

Outputs

The prediction results shall include:

- Customer ID
- Prediction Result
- Churn Probability
- Risk Score
- Risk Category
- Retention Recommendation

Acceptance Criteria

The feature shall be considered complete when:

- Users can upload a valid CSV file.
- Invalid files are rejected with appropriate validation messages.
- Predictions are generated successfully for all valid records.
- Results can be downloaded successfully.

Feature FTR-009

Report Generation

Description

The Report Generation feature enables users to generate business reports containing customer churn insights, prediction summaries, customer rankings, segmentation information, and other analytical results.

Reports are intended to support management review, operational decision-making, and business performance analysis.

The generated reports shall be available for download in commonly used document formats.

Business Value

Business stakeholders require periodic reports for review meetings, performance evaluation, and strategic planning.

Automated report generation eliminates manual preparation and ensures consistent reporting across the organization.

Functional Capabilities

The system shall generate reports containing:

- Customer Summary
- Prediction Summary
- High-Risk Customers
- Customer Ranking
- Customer Segmentation
- Dashboard Summary
- Analytics Summary

Export Formats

The system shall support report export in:

- PDF
- CSV
- Microsoft Excel (.xlsx)

Acceptance Criteria

The feature shall be considered complete when:

- Reports are generated successfully.
- Reports contain accurate business information.
- Reports can be exported in supported formats.
- Generated reports maintain a professional format suitable for business review.

Feature FTR-010

Model Performance Dashboard

Description

The Model Performance Dashboard provides an overview of the machine learning model's performance and prediction reliability.

The dashboard is intended primarily for administrators and technical reviewers to evaluate the effectiveness of the prediction model.

This feature promotes transparency by presenting commonly accepted machine learning evaluation metrics.

Business Value

Although business users are primarily interested in prediction results, understanding model performance increases confidence in the system and helps validate that the prediction model is performing as expected.

The dashboard also assists future model improvements by providing evaluation metrics.

Functional Capabilities

The dashboard shall display:

- Overall Model Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix
- Feature Importance

Model Insights

The dashboard shall also display:

- Most Important Prediction Factors
- Feature Importance Ranking
- Overall Model Performance Summary

Acceptance Criteria

The feature shall be considered complete when:

- Model evaluation metrics are displayed correctly.
- Performance metrics are calculated accurately.
- Feature importance information is available.
- The dashboard presents information in a clear and understandable manner.

9. Functional Requirements

9.1 Overview

Functional requirements define the behavior that the Subscription Cancellation Prediction System must provide in order to satisfy business objectives. These requirements describe the expected functionality of the application from the user's perspective.

FR-001 User Authentication

The system shall provide secure authentication for authorized users before granting access to the application.

FR-002 Dashboard

The system shall display an executive dashboard summarizing customer churn statistics and key business metrics.

FR-003 Customer Management

The system shall allow users to view customer information through a searchable and filterable customer list.

FR-004 Customer Profile

The system shall display a detailed customer profile containing customer information, prediction results, risk score, prediction explanation, and retention recommendations.

FR-005 Churn Prediction

The system shall predict whether a customer is likely to cancel their subscription using the trained machine learning model.

FR-006 Customer Ranking

The system shall rank customers according to their churn probability.

FR-007 Customer Segmentation

The system shall group customers into business-relevant segments based on customer characteristics.

FR-008 Explainable AI

The system shall display the primary factors influencing each churn prediction.

FR-009 Retention Recommendation

The system shall recommend appropriate retention strategies based on customer risk.

FR-010 Dashboard KPIs

The system shall display business KPIs related to customer churn and customer behavior.

FR-011 Analytics

The system shall provide interactive charts and analytical visualizations for business analysis.

FR-012 Search & Filter

The system shall allow users to search and filter customer records using supported criteria.

FR-013 Bulk Prediction

The system shall allow prediction of multiple customer records through CSV upload.

FR-014 Report Generation

The system shall generate downloadable reports containing customer predictions and analytical summaries.

FR-015 Model Performance

The system shall display machine learning evaluation metrics to provide transparency regarding model performance.

10. Non-Functional Requirements**10.1 Overview**

Non-functional requirements define the quality attributes and operational characteristics of the Subscription Cancellation Prediction System. While functional requirements describe what the system should do, non-functional requirements specify how well the system should perform under expected operating conditions. The system shall satisfy the following quality requirements throughout development and deployment.

10.2 Performance

The system shall provide a responsive and efficient user experience.

NFR-001

The application shall load the dashboard within **3 seconds** under normal operating conditions.

NFR-002

Single customer churn prediction shall complete within **2 seconds**.

NFR-003

Bulk prediction shall process uploaded datasets efficiently without system failure.

NFR-004

Search and filtering operations shall return results with minimal delay.

10.3 Security

The application shall protect business data from unauthorized access.

NFR-005

Only authenticated users shall access the system.

NFR-006

Passwords shall be securely stored using industry-standard hashing techniques.

NFR-007

Role-based authorization shall be supported to restrict access to application features.

NFR-008

The application shall validate all user inputs before processing.

10.4 Availability**NFR-009**

The application shall remain available during normal business hours with minimal downtime.

10.5 Scalability**NFR-010**

The application shall support future expansion to accommodate larger customer datasets and additional business features without requiring significant architectural changes.

10.6 Usability**NFR-011**

The application shall provide an intuitive and user-friendly interface requiring minimal training.

NFR-012

Business information shall be presented using simple language and clear visualizations.

10.7 Reliability**NFR-013**

Prediction results shall remain consistent when identical input data is provided.

NFR-014

The application shall handle invalid inputs gracefully by displaying meaningful validation messages.

10.8 Maintainability**NFR-015**

The system shall be designed using a modular architecture to simplify future maintenance and feature enhancements.

11. Dataset Overview**11.1 Overview**

The Subscription Cancellation Prediction System will utilize a structured customer dataset containing historical subscription and customer behavior information. The dataset has been provided by the business owner for developing and evaluating the churn prediction model.

The dataset is intended for supervised machine learning, where the target variable indicates whether a customer is expected to cancel their subscription within the next three months.

11.2 Dataset Summary

Attribute	Value
Dataset Type	Structured Tabular Dataset
Number of Records	15,946
Number of Features	14
Target Variable	Will_Cancel_Next_3_Months
Prediction Type	Binary Classification
Missing Values	None

11.3 Target Variable

The prediction model will use the following target variable:

Will_Cancel_Next_3_Months

Possible values:

- **0** – Customer is not expected to cancel the subscription.
- **1** – Customer is expected to cancel the subscription.

This target variable forms the basis of the supervised learning process and is used to train and evaluate the prediction model.

11.4 Available Dataset Features

The dataset contains customer demographic, subscription, usage, and behavioral information.

The available features are:

- Customer_ID
- Age
- Income_Level
- Number_of_Subscriptions
- Tenure_Months
- Monthly_Total_Spend
- Avg_Usage_Hours_Per_Week
- App_Switch_Frequency
- Customer_Support_Interactions
- Satisfaction_Score
- Discount_Used
- Device_Type
- Payment_Mode
- Will_Cancel_Next_3_Months (Target)

These features provide sufficient information for developing a predictive churn model and generating business insights.

12. Dataset Analysis

12.1 Data Quality Assessment

A preliminary analysis of the provided dataset indicates that the data is suitable for machine learning model development.

The dataset demonstrates the following characteristics:

- The dataset contains no missing values.
- The target variable is clearly defined.
- Both numerical and categorical features are available.
- Customer behavioral attributes are included.
- Subscription-related information is available.
- Customer satisfaction information is available.

Overall, the dataset is considered clean and ready for preprocessing and model development.

12.2 Feature Categories

For analysis purposes, the available dataset attributes can be grouped into the following categories.

Customer Information

- Customer ID
- Age
- Income Level

Subscription Information

- Number of Subscriptions
- Tenure Months
- Monthly Total Spend
- Discount Used

Customer Behaviour

- Average Usage Hours Per Week
- App Switch Frequency
- Customer Support Interactions
- Satisfaction Score

Technical Information

- Device Type
- Payment Mode
-

Prediction Label

- Will_Cancel_Next_3_Months

12.3 Business Value of Available Features

The available dataset provides sufficient information to support several business objectives.

Examples include:

- Predicting customer churn.
- Identifying high-risk customers.
- Ranking customers by churn probability.
- Segmenting customers into meaningful business groups.
- Generating customer analytics.
- Providing explainable AI insights.
- Supporting customer retention strategies through business rules.

13. Dataset Limitations

13.1 Overview

Although the provided dataset is suitable for building the proposed churn prediction system, it does not contain all customer attributes typically available in enterprise subscription management platforms.

As a result, certain business scenarios cannot be fully addressed within the scope of the current dataset.

The following limitations have been identified during dataset analysis.

13.2 Identified Limitations

The current dataset does not include information related to:

- Payment failure history.
- Subscription renewal dates.
- Auto-renewal status.
- Customer login history.
- Last customer activity date.
- Refund requests.
- Historical monthly customer activity.
- Customer cancellation reasons.

13.3 Impact on the Product

Due to the absence of the above information, the following capabilities cannot be fully implemented:

- Analysis of payment-related churn.
- Identification of inactive customers based on login history.
- Renewal-based churn analysis.
- Refund behavior analysis.
- Time-based customer activity analysis.

Despite these limitations, the available dataset supports the successful implementation of the core product features, including churn prediction, customer profiling, customer ranking, segmentation, explainable AI, retention recommendations, dashboards, analytics, bulk prediction, report generation, and model performance evaluation.

14. Assumptions

The following assumptions have been considered during the preparation of this Product Requirements Document. These assumptions are intended to guide project planning and development activities.

AS-001

The provided customer dataset accurately represents customer behavior and subscription information.

AS-002

The target variable (Will_Cancel_Next_3_Months) is correctly labeled and suitable for supervised machine learning.

AS-003

The dataset used during development is representative of the business problem and sufficient for building the initial version of the prediction model.

AS-004

The application will initially support a single user role (Administrator) with complete access to all modules.

AS-005

The churn prediction model will be trained offline using the provided historical dataset before deployment.

AS-006

Business users possess basic knowledge of customer management and business analytics.

AS-007

The application will be used as an internal business tool and will not be directly accessible to customers.

15. Constraints

The project is subject to the following constraints.

CN-001

The prediction model will rely solely on the information available in the provided dataset.

CN-002

The application will not perform real-time customer activity monitoring.

CN-003

Prediction quality depends on the quality and completeness of the provided dataset.

CN-004

Customer retention recommendations will be based on predefined business rules in the current version.

CN-005

The project is intended as a web-based application and does not include a mobile application.

CN-006

Future business requirements may require additional datasets and further model training.

16. Proposed Technology Stack (Recommendation)

The following technologies are recommended for implementing the Subscription Cancellation Prediction System. Final technology selection and implementation details remain subject to approval by the Technical Team Lead.

Layer	Recommended Technology
Frontend	React.js
UI Components	Material UI
Data Visualization	Chart.js / Recharts

Layer	Recommended Technology
Backend	FastAPI (Python)
Machine Learning	Scikit-learn, CatBoost, SHAP
Data Processing	Pandas, NumPy
Database	PostgreSQL
Authentication	JWT
Version Control	Git & GitHub
Deployment	Docker (Optional)

17. Project Deliverables

Upon successful completion of the project, the following deliverables are expected.

Software Deliverables

- Subscription Cancellation Prediction Web Application
- Machine Learning Prediction Model
- Customer Management Module
- Dashboard Module
- Analytics Dashboard
- Report Generation Module
- Bulk Prediction Module
- Model Performance Dashboard

Documentation Deliverables

- Product Requirements Document (PRD)
- Technical Design Document (Prepared by Technical Team Lead)
- Source Code Documentation
- User Guide
- Installation Guide

Business Deliverables

- Churn Prediction Results
- Customer Risk Scores
- Customer Ranking
- Customer Segmentation
- Business Reports
- Customer Analytics

18. Proposed Sprint Plan

The following sprint plan is proposed to guide the development process. The final sprint schedule and task allocation shall be determined by the Technical Team Lead.

Sprint	Date	Objectives	Deliverables
Sprint 1	02-07-2026	Requirement finalization, project planning, dataset analysis, UI planning	Approved requirements, UI wireframes, project setup
Sprint 2	05-07-2026	Core development including authentication, customer management, and ML model	Customer management module and trained

Sprint	Date	Objectives	Deliverables
		development	prediction model
Sprint 3	08-07-2026	Churn prediction, customer profile, risk score, explainability, ranking, segmentation, and retention recommendations	Intelligent prediction module
Sprint 4	11-07-2026	Dashboard, analytics, customer search, filtering, bulk prediction, and report generation	Analytics dashboard and reporting module
Sprint 5	14-07-2026	Model evaluation, testing, bug fixing, deployment preparation, documentation, and final demonstration	Final integrated application

19. Acceptance Criteria

The project shall be considered complete when the following acceptance criteria have been satisfied.

AC-001

The application successfully predicts customer churn using the provided dataset.

AC-002

The system displays a churn risk score for every customer.

AC-003

Customer profiles display prediction results, customer information, risk score, and prediction explanation.

AC-004

Customers are ranked according to churn probability.

AC-005

Customer segmentation is successfully generated and displayed.

AC-006

Retention recommendations are displayed for customers identified as high risk.

AC-007

The executive dashboard displays all required business KPIs.

AC-008

Interactive analytics are available and display customer insights correctly.

AC-009

Customer search and filtering operate correctly.

AC-010

Bulk prediction successfully processes uploaded customer datasets.

AC-011

Business reports are generated successfully in supported formats.

AC-012

The Model Performance Dashboard displays evaluation metrics and feature importance.

20. Project Risks

The following risks have been identified during project planning.

Risk	Impact	Mitigation
Limited dataset attributes	Medium	Clearly define supported features and document limitations
Prediction accuracy lower than expected	Medium	Evaluate multiple ML models and select the best-performing one
Changes in business requirements	Medium	Prioritize requirements and manage scope through sprint planning
Integration challenges	Low	Follow modular development practices
Project timeline constraints	Medium	Prioritize must-have features for initial release

21. Future Scope

The current project establishes the foundation for a customer churn analytics platform. Future releases may introduce additional capabilities to further improve customer retention and business intelligence.

Potential future enhancements include:

- Real-time customer activity monitoring.
- Automated model retraining.
- Customer Lifetime Value (CLV) prediction.
- CRM integration.
- Email and notification services.
- Personalized recommendation engine using advanced machine learning.
- AI-powered business assistant for generating analytical summaries.
- Multi-role access with Business Analyst and Customer Success Executive dashboards.
- Cloud deployment for enterprise-scale usage.
- Time-series analysis for customer behavior trends.

22. Conclusion

The Subscription Cancellation Prediction System is designed to provide businesses with an intelligent, data-driven solution for identifying customers who are likely to cancel their subscriptions. By combining machine learning, customer analytics, explainable AI, and business reporting within a unified web application, the proposed system aims to improve customer retention, reduce revenue loss, and support informed business decision-making.

The provided dataset supports the implementation of the core product features, including churn prediction, customer profiling, risk assessment, customer ranking, segmentation, retention recommendations, dashboards, analytics, bulk prediction,

report generation, and model performance evaluation. While certain advanced business scenarios are constrained by the available dataset, the current scope is sufficient to deliver a valuable and functional product.

This Product Requirements Document serves as the official business handover to the Technical Team Lead. It defines the product vision, scope, requirements, features, and expected deliverables while allowing the technical team to determine the most appropriate implementation approach, architecture, and development strategy.